Ethics and Regulations in Explainable AI

Identifying Ethical Risks in Opaque AI Systems

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Globomantics Storyline

AI systems

Expansion

Regulatory challenges

Case Study: Tech Giant AI Recruiting

AI ranks CVs

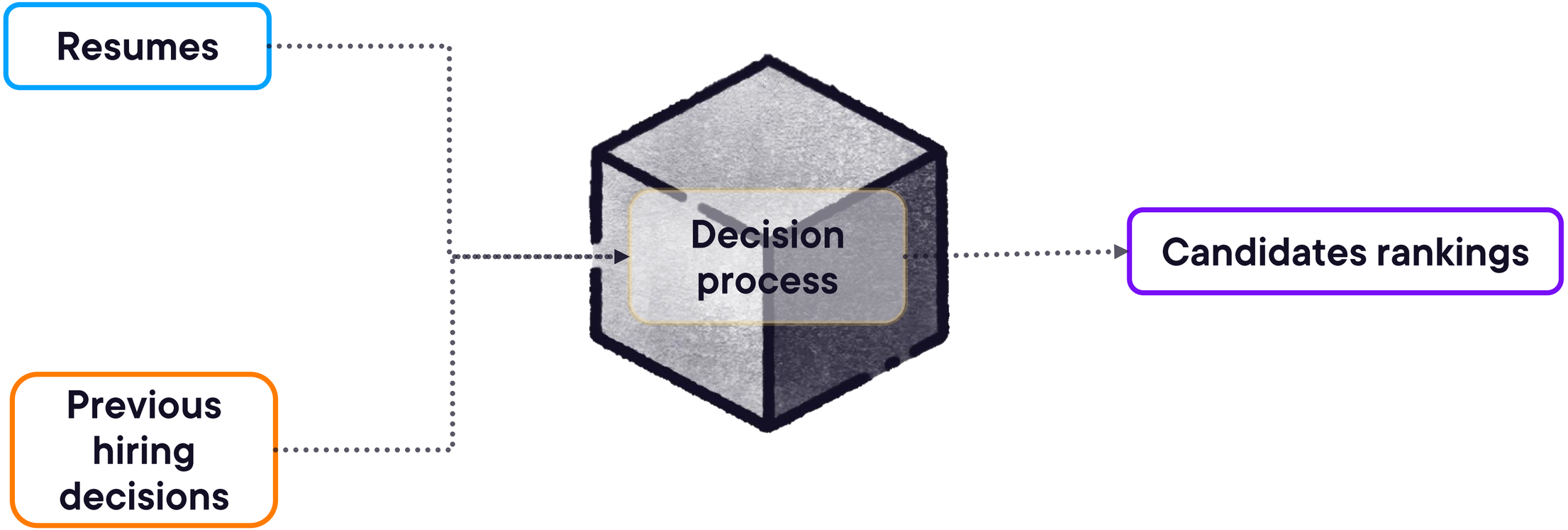
Systematic discrimination

How Did This Happen?

Historical reality	vs.	What AI learned
Tech industry mostly male		Male characteristics predict success
Complex factors which resulted in fewer women in tech		Being a woman as negative indicator for job performance
Educational pipeline issues		Amplify discrimination

Black Box AI

Black Box AI


Resumes
Previous hiring decisions
Decision process
Candidates rankings

Consequences

Human

Business

Legal

Why Explainability Matters

Early detection

Understand rationale of decisions

Trust

Regulatory compliance

More Ethical Risks

Racial

Socioeconomic

Age

Disability

Geographic

The Dark Side of Black Box AI

What Types of Harm?

Discriminatory

Quality of service

Procedural

Autonomy

Case Study: Recidivism Risk Algorithm

US criminal justice system

Assess likelihood of reoffending

High-stakes decisions:

- Sentencing
- Bail
- Parole

Proprietary algorithm

Consequences of Poor Explainability

Judges don't know how scores are calculated

Defendants have few opportunities to challenge scores

Researchers struggle to audit

"The formula was particularly likely to falsely flag black defendants as future criminals, wrongly labeling them this way at almost twice the rate of white defendants."

Citation: ProPublica, Machine Bias, 2016, https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

Algorithmic Harm

Discriminatory
Systematic racial bias

Quality of service
Low accuracy

Procedural
Difficult to challenge scores

Autonomy
Judges get influenced

Key Points

Need for explainability

Prevent harm

Demo: Essential Approaches for Ethical AI

Essential Approaches

Model cards

Info on AI models

Datasheets

Info on AI datasets

Key Sections of Model Cards

Model details

Intended use

Quantitative analysis

Ethical considerations

Model Details

model_details.yaml

```yaml
developer: "Globomantics Financial AI Team"
model_date: "March 2024"
model_version: "v3.1"
model_type: "Gradient Boosting Classifier"

contact: "financial-ai@globomantics.com"

regulatory_compliance:
  - "Fair Credit Reporting Act"
  - "GDPR Article 15"
```

Intended Use

intended_use.yaml

```yaml
primary_intended_uses:
  - "Initial credit risk assessment for personal loans"
  - "Automated pre-screening for loan applications"

target_users:
  - "Bank loan officers and credit analysts"
  - "Automated lending platforms"

explicitly_out_of_scope:
  - "Final loan approval decisions (human required)"
  - "Business or commercial loan assessment"
```

Quantitative Analysis

quantitative_analysis.yaml

```yaml
performance_by_age:
  age_25_35:
    precision: 0.89
    recall: 0.85
  age_51_65:
    precision: 0.82
    recall: 0.79
  age_65_plus:
    precision: 0.78
    recall: 0.74
```

Ethical Considerations

ethical_considerations.yaml

```yaml
ethical_risks:
  - impact: "Access to credit and financial opportunity"
    risk_level: "high"
    affected_groups: ["old applicants"]

safeguards:
  - "Protected characteristic monitoring (race, age, ethnicity)"
  - "Human review for decisions below confidence threshold"
  - "Regular fairness audits"
```

Key Sections of Datasheets for Datasets

Composition

Collection process

Uses

Composition

composition.yaml

```yaml
composition:
  total_instances: 75000
  time_period: "2019-2023 loan decisions"

  representation_gaps:
    - "Only 3% Black applicants"
    - "65% above-median income skews toward higher earners"
```

Collection Process

collection_process.yaml

```yaml
collection_process:
  collection_mechanism: "Automated system during application process"

  data_sources:
    primary: "Internal loan application system"
    secondary:
      - "Credit bureau reports"
      - "Bank account history"

  known_collection_biases:
    - "Bank branch locations concentrated in urban areas"
    - "Online application process may exclude lower-tech demographics"
```

Uses

uses.yaml

```yaml
uses:
  current_uses:
    - "Personal loan risk assessment models"
    - "Credit scoring algorithm training"

  inappropriate_uses:
    - "Mortgage lending (different regulatory requirements)"
```

Resources Referenced in This Module

https://www.reuters.com/article/us-amazon-com-jobs-automation-insight-idUSKCN1MK08G/, 2018

Machine Bias, https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing, 2016

Mitchell et al., Model Cards for Model Reporting, 2019
https://arxiv.org/abs/1810.03993

Gebru et al., Datasheets for Datasets, 2021,
https://arxiv.org/pdf/1803.09010